

SANDIPAN GHOSH

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PROFESSIONAL SUMMARY

Senior Software Architect with **21+ years** of EDA and physical design experience at Cadence Design Systems, including **7+ years as lead architect of device auto-placement and routing** infrastructure. Central designer of a patented **DRC-Aware Routing Grids System** (US 11,790,147) that decouples routing algorithms from PDK-specific DRC rules, cutting new-node enablement from months to weeks across TSMC N3/N5/N7, Intel 1278, and Samsung nodes. Architect of "Golden" layout engines deployed at Apple, NVIDIA, Intel, Samsung, TSMC, and Qualcomm. Now extending these deterministic engines with **Agentic AI** for fully autonomous memory chip assembly routing -- a production-scale C++ engine orchestrated by LangGraph targeting 1M+ net capacity.

What makes me different: I am one of the few engineers in the industry who has both **designed and shipped production routing infrastructure** at Cadence for 7+ years **and** built production-grade Agentic AI systems that can wrap these engines for autonomous design closure -- not demos, not prototypes, but fully-specified engines with deterministic verification and pluggable EDA tool integration.

DEVICE AUTO-PLACEMENT & ROUTING -- 7+ Years as Lead Architect

DRC-Aware Routing Grids System (Patented -- US 11,790,147)

Central architect of Cadence's **Grid-Based Device Router (GBR)** -- a PDK-independent routing system for Virtuoso that separates DRC rule encoding from routing algorithm logic:

- **Routing Graph Architecture:** Designed a dynamic graph that is a virtual representation of all available routing space across all metal layers. Edges are classified as free, occupied, or haloed. Graph updates in real-time as routing segments and vias are created. Width-Spacing Pattern (WSP) tracks form the foundation; pre-routed shapes and route segments become occupied edges; DRC influence zones become halo edges.
- **Segment Service Module:** Computes shadows/halos for routed segments from cached DRC constraints (minArea, minLength, minSpacing, EOL, minSpanLengthSpacing). Returns shadow boxes for short segments and separate end-of-segment shadows for segments exceeding minimum length constraints.
- **Via Service Module:** Provides best via selection for each layer/width pair and computes clearance shadows/halos (same-mask, diff-mask, same-net, diff-net) that guarantee DRC correctness with other potential shapes.
- **Interlayer Edge System:** Intersections between different-layer tracks are treated as potential via locations with free/occupied status tracking.
- **PDK Enablement Results:** TSMC N7 (3 months), N5 (2 months), N3 (1 month), Cadence GPDK (2 weeks) -- each successive node faster due to architecture scalability. The GBR abstracts all DRC rules into the graph, keeping the routing engine PDK-independent. DRC/LVS signoff compliance is verified incrementally as each segment and via is placed.
- **Multi-Net Resource Allocation:** Addressed the fundamental congestion problem where greedy net-sequential routing causes early-routed nets to occupy all lower-metal resources, blocking subsequent nets.

Bayesian Optimization of Router Parameters (Internal R&D)

Led exploration of **Optuna-based hyperparameter optimization** (Python) applied to the GBR, treating the router as a black box:

- **Framework:** Virtuoso Studio calls Python Flask REST API, which invokes Python/Optuna. Optuna suggests trial routing parameters (preferred layer settings, via/metal costs, net ordering); routed models are evaluated and scores fed back for Bayesian incorporation.
- **Congestion Reduction:** Optimizer discovers parameters that dissuade the router from greedily occupying lower-metal layers, instead encouraging topologies that escape to higher, vertical layers -- reducing resource contention for multi-net routing.
- **Resistance Optimization:** Optimizer steers routing toward less-resistive higher metal layers instead of highly-resistive lower metals (M0), resulting in 57% reduction in highly-resistive metal usage and 352-627% increase in moderately/less resistive metal usage.
- **Results:** Reduced opens from 25 to 3 on representative testcases (score from 28M to 3M). Validated on TSMC N3 (M0 on/off), Intel 1278. 100 trials on 20 parallel threads, optimal parameters at trial 60-80. Can run in 5x single-route duration.
- **Proposed Solutions:** Warm-start (reuse pre-trained parameters across similar PDK/designs without re-training) and one-click autonomous routing (allocate training duration, walk away, get optimal routing).

Device Fill, Placement & Infrastructure

- **Metal & Poly Filler Engines:** Architected Virtuoso's density fill engines -- antenna fixing (jumpers, diodes), layer density analysis, DFM compliance for advanced nodes.
- **SADP Grid Infrastructure:** Grid models, checkers, and tools for Self-Aligned Double Patterning at 10nm/7nm -- enabling placement and routing readiness.
- **Width-Spacing Patterns** (US 10,733,351): Automatic WSP generation for color-based multi-patterning track systems by examining instance heights, placement orientations, pin widths/spacing/colors, and handling flipped/mirrored instances.
- **Automatic Placement/Routing Infrastructure Generation:** Developed infrastructure for automatic device placement and routing on advanced nodes.

AGENTIC AI: Autonomous Memory Chip Assembly Routing Engine

Developed a **production-scale agentic framework** that automates complex VLSI routing orchestration for memory chip assembly (DRAM/SRAM/Flash) -- replacing manual design closure with agent-driven, mathematically verified automation.

Architecture (10,000+ lines of specification-complete design documents)

Component	Description	Scale
Memory Routing Architecture v4.8	Domain architecture for memory chip assembly routing -- Block (array internals: bitlines, wordlines, sense amps) vs Periphery (fishbone topology: spine, ribs, tap connections). Two distinct routing domains sharing algorithms and verification.	2,155 lines

C++ Implementation Spec v2.0	RouterBase -> BlockRouter + PeripheryRouter hierarchy. RoutingToolPool with 14 stateful function objects. EDA callbacks for Virtuoso, Innovus, IC Compiler II, OpenROAD. Database adapters for OpenAccess, Catena, CSV.	2,261 lines
Pin-to-Spine Tool (P2S) v1.0	Standalone libp2s. a static library + CLI binary. 39 individually-callable function objects in 7-phase pipeline. Bidirectional track model (mature node). Zero-shorts hard constraint. extern "C" API. CSV/OpenAccess/Catena adapters.	1,650 lines
Pin-to-Spine QoR Deep Dive	Failure mode analysis: 12 open modes, 8 short modes, 14 DRC violation categories. 39-feature evaluation rubric (PA-1 through RD-6). 13 identified gaps in Virtuoso's existing implementation. 9 established industry techniques documented.	999 lines
LangGraph Orchestrator v3.0	AI brain -- topology selection (H-tree vs Fishbone based on die aspect ratio), solver dispatch, DRC/LVS verification. Three-layer stack: Orchestrator + Solver Engine + Domain Workers. Six routing domain taxonomy.	940 lines
Meta-Optimizer (AMOF)	LLM-guided multi-objective optimization framework. 5 solver pool (Gurobi, SciPy, Optuna, DEAP, OR-Tools). Completeness gate ensures no missing constraints before solving. GA chromosome design for routing. MCP server interface.	1,865 lines
VLSI Routing Topologies	H-tree vs Fishbone routing analysis. Die geometry impact on clock/power/signal distribution. Transition zones. Hybrid topologies. DRAM internals: destructive read, charge sharing, folded bitline, sense amp drain, memory banking.	1,928 lines

Key Design Decisions

- **Hybrid Architecture:** Python/LangGraph orchestrates strategy (algorithm selection, net ordering, solver switching); C++ engine executes at microsecond-per-edge speed. Built-in DRC/LVS verification at every routing step. Zero hallucination in the compute path.
- **Zero-Shorts Hard Constraint:** If a connection cannot be made without creating a short, leave it OPEN. An open is a known failure. A short is silent corruption.
- **Database Agnostic:** Pluggable adapter interface -- CSV (testing), OpenAccess (Cadence), Catena (IBM). Core engine has no external runtime dependency.
- **Capacity:** 1,000,000 nets targeting Samsung / SK Hynix / Micron memory scale across all die aspect ratios.

PROFESSIONAL EXPERIENCE

Cadence Design Systems -- 21 years 6 months (October 2004 - Present)

Senior Software Architect (July 2024 - Present) | San Francisco Bay Area

- **Agentic AI + Device Routing:** Architecting the integration of Python-based frameworks (LangGraph, Crew AI, MCP Servers, AutoGen 2, OpenAI Agents SDK) with production device routing and fill tools for autonomous design closure.
- **Autonomous Memory Routing Engine:** Designed a C++ memory routing engine with LangGraph orchestration targeting advanced nodes (TSMC N2, Intel 18A) for memory chip assembly. Produced 10,000+ lines of architecture specifications across 7 interconnected documents.
- **Pin-to-Spine Tool:** Specified a standalone routing library (`libp2s.a`) with 39 function objects, 7-phase pipeline, bidirectional track model, and zero-shorts hard constraint.
- **Optimization Frameworks:** Implemented 20+ Python-based optimization solutions using Gurobi, SciPy, PuLP, OR-Tools, DEAP (Genetic Algorithms), and Optuna for routing, placement, and resource allocation problems.

Software Architect (August 2023 - July 2024) | San Jose, CA

- Continued evolution of the DRC-Aware Routing Grids System and density filling solutions for advanced process nodes.
- Extended device routing architecture for multi-foundry support across TSMC, Intel, and Samsung PDKs.

Software Architect (July 2022 - August 2023) | Montreal, Quebec, Canada

- Led Bayesian Optimization exploration for router parameter tuning -- Python/Optuna-based hyperparameter optimization treating the GBR as a black box, achieving 57% reduction in resistive metal usage and reducing opens from 25 to 3.

Software Architect (October 2020 - July 2022) | San Francisco Bay Area

- **Led 5-engineer team** architecting the **DRC-Aware Routing Grids System** (US 11,790,147 -- granted Nov 2023) -- a novel routing graph architecture that decouples PDK-specific DRC rules from core routing algorithms.
- **Routing Graph Design:** Dynamic graph representing all available routing space across all metal layers. Edges encode DRC rules via Segment Service and Via Service modules. Graph updates in real-time as segments and vias are created.
- **PDK Enablement Acceleration:** TSMC N7 (3 months), N5 (2 months), N3 (1 month), GPDK (2 weeks) -- each successive node faster due to architecture scalability.
- Developed density filling and automatic placement/routing infrastructure generation solutions for 5nm and 3nm nodes.
- Designed modular C++ and SKILL-based architecture for high performance with fast-changing requirements.

Senior Member of Consulting Staff / R&D Engineering Manager (November 2019 - October 2020) | San Jose

- Managed team of 5 engineers developing a fast device routing solution.
- Direct technical oversight of routing engine performance, quality, and advanced node enablement.

Senior Member of Consulting Staff -- Individual Contributor (February 2016 - November 2019) | San Jose

- **Metal & Poly Filler Engines:** Primary developer for Virtuoso's metal and poly layer filler engines, including antenna fixing modules (jumpers, diodes) and layer density analysis -- critical infrastructure for device routing quality.

- **SADP Grid Infrastructure:** Developed grid models, checkers, and helper tools for Self-Aligned Double Patterning processes -- enabling layout readiness for placement and routing at 10nm and 7nm nodes.
- **Width-Spacing Patterns** (US 10,733,351): Automatic WSP generation for color-based multi-patterning track systems -- the foundation upon which the routing graph is built.
- **Technology Stack:** C++, STL, Boost, Qt-based infrastructure.

Senior Member of Consulting Staff (July 2013 - February 2016) | Noida, India

Led development of major features in **Virtuoso Layout Editor** (>70% market share, industry-standard custom layout tool):

- **FinFET Support:** Implemented advanced node (16nm and below) editing capabilities for finFET devices.
- **Neighborhood Awareness:** Designed and built context-sensitive editing commands with aperture-based layout analysis (patented).
- **Motif-to-Qt Migration:** Led the transition of the legacy Motif-based GUI to Qt framework.
- **New Features:** Layer Palette, Display Resource Editor, Fast Alignment Command, Smart Ruler, Fluid Guard-ring Support.
- **Awards:** Won 2 prestigious organizational awards. 3 patents granted (US 9305133, US 8612923, US 8271909).

Member of Consulting Staff (July 2010 - July 2013) | Noida, India

- **Constraint GUI:** Designed and built from scratch a Qt-based constraint editor supporting wide-range GUI operations for design and foundry constraints on selected objects. Won **Organization Level Award for Execution**.
- **Patent Granted:** 1 patent for constraint editing system.
- **Conference Paper:** Authored and presented paper at Cadence Technical Conference (US).
- Transitioned to Virtuoso Layout Editor -- worked across almost all sub-tools.

Senior Member Technical Staff (April 2007 - July 2010) | Noida, India

- **Translators:** Primary developer for LEF-DEF and GDS/Stream translators -- critical tools for design format interchange.
- **Performance & Quality:** Focused on coverage, Purify, and runtime performance optimization.
- **Awards:** Won 2 departmental awards.
- **Conference Paper:** Authored and presented paper at Cadence Technical Conference (US).

Member Technical Staff (October 2004 - April 2007) | Noida, India

- Early career development in EDA infrastructure and tool development.

Mentor Graphics -- Associate Member Technical Staff (October 2003 - October 2004) | Hyderabad, India

- **Primary lead** for porting DxDesigner front-end to Linux using Mainwin (Mainsoft) -- providing MFC, COM runtime, and Windows registry equivalents on Unix.
- Served as primary contact for all Mainwin-related communications with Mainsoft engineering support across multiple Unix platforms.

Geometric Software -- Software Engineer (August 2002 - October 2003) | Pune, India

- Worked on CATIA CAD designer tool and ENOVIA PLM solutions from Dassault Systems.

AGENTIC AI PROJECT PORTFOLIO

EDA / Silicon Projects

#	Project	Framework	Description
1	VLSI Memory Router	LangGraph + C++	Autonomous routing engine for DRAM/SRAM/Flash -- 7 interconnected spec documents, 10,000+ lines
2	Pin-to-Spine Tool (P2S)	C++, extern "C"	Standalone libp2s.a with 39 function objects, zero-shorts constraint, bidirectional tracks
3	VLSI Routing (DEAP + A*)	DEAP (GA), rustworkx	Multi-layer VLSI grid routing with genetic algorithm net ordering
4	ModGen Scaling (RSMT)	NetworkX, Steiner trees	Rectilinear Steiner minimum tree for module generation

Other Agentic AI Projects

#	Project	Framework	Description
5	Meta-Optimizer (AMOF)	LangGraph	LLM-guided solver selection with Gurobi/SciPy/DEAP/OR-Tools/Optuna pool and completeness gate
6	Chat Assistant	OpenAI Agents SDK	Multi-provider LLM (Gemini primary, GPT-4 fallback) orchestrating 14 MCP servers with rate-limiting and retry
7	MedExplainer	LangGraph	Medication journey narrator with 7 medical API sources, lab OCR, prescription OCR. 239 tests passing.
8	College & Career Planner	LangGraph, 6 MCP servers	AI-resilience scoring, RIASEC profiling, 7 specialized agents for student guidance
9	Stock Picker	Crew AI	Multi-agent stock analysis with sector filtering, S&P 500 metadata, yfinance integration
10	Portfolio Manager	Crew AI	Automated portfolio analysis and rebalancing recommendations
11	Web Scraper	AutoGen 2, MCP Fetch	Smart input parsing with optional recursive crawl and content extraction

MCP Servers (Custom Built)

Server	Capability
E*TRADE MCP	Full brokerage integration -- accounts, portfolios, quotes, option chains, order placement via OAuth2
Gmail MCP	Read, search, send emails via Gmail API with OAuth2
Outlook MCP	Microsoft Outlook desktop integration for work email
Pushover MCP	Mobile push notifications with emergency escalation
MCP Proxy	Security sandbox -- intercepts and validates MCP tool calls between LLM and backend servers

Optimization & Algorithm Implementations

Implementation	Solver/Library	Domain
Facility Location Optimization	Gurobi	Capacitated facility placement
Multi-Objective Optimization	Gurobi	Weighted-sum scalarization with Pareto frontier
Electric Car Profit Maximization	SciPy	Nonlinear constrained optimization
N-Queens (3 approaches)	PuLP, OR-Tools, DEAP	LP, constraint programming, genetic algorithm comparison
Resource Assignment	PuLP	Integer programming for team allocation
Network Minimum-Cost Flow	NetworkX	Graph-based transportation optimization
Police Scheduling	PuLP	Shift scheduling with fairness constraints
Symbolic Regression	DEAP (GP)	Genetic programming for equation discovery
Travel Planning	OR-Tools	Vehicle routing with time windows

TECHNICAL PROFICIENCIES

Domain	Technologies
Device Routing	Grid-Based Router (GBR), DRC-aware routing graphs, net ordering, congestion management, Segment/Via Services, PDK enablement
Agentic AI Frameworks	LangGraph, Crew AI, AutoGen 2, OpenAI Agents SDK, MCP (Model Context Protocol) Architecture
LLM Integration	Google Gemini (2.5 Flash/Pro), GPT-4.1, Claude (Code/Sonnet), multi-provider fallback chains
EDA / Physical Design	Virtuoso Layout Editor, Auto P&R, SKILL, OpenAccess API, DFM, LEF/DEF/GDSII translators
Process Technology	finFET, gaaFET PDKs -- TSMC (A14/16, N2/N2P/N3/N4/N5/N6/N7/N12/N16), Intel (18A/20A), Samsung (SF2/3/4/8)
Optimization / Solvers	Gurobi, SciPy, PuLP, OR-Tools, DEAP (Genetic Algorithms), Optuna (Bayesian)
Graph Algorithms	A*, Steiner Tree (RSMT), MST, KD-Tree, Network Flow, Two-Level GA, Pattern Router
Languages	C++, Python 3.x, Cadence SKILL, SQL
C++ Infrastructure	STL, Boost, Qt, CMake, pybind11, GoogleTest
DevOps / Tools	Git, Docker, CI/CD, VS Code (Agentic modes), Cursor, Claude Code

PATENTS (7 Granted)

- System and method for routing in an electronic design** (US 11,790,147 -- Nov 2023) -- Dynamic routing graph generation from design rules, representing all available routing space across all layers, with real-time graph updates as routing segments and vias are created.
- Method, system, and computer program product for rearrangement of objects within an electronic design** (US 10,769,346 -- Sep 2020) -- Automated layout object rearrangement: dragged objects trigger automatic repositioning of existing objects to resolve spacing violations and overlaps.
- Generating width spacing patterns for multiple patterning processes** (US 10,733,351 -- Aug 2020) -- Automatic creation of width-spacing patterns (WSPs) for color-based track systems by examining instance heights, placement orientations, pin widths/spacing/colors, and handling flipped/mirrored instances.
- System and method for selective application and reconciliation of hierarchical ordered sets of circuit design constraints within a circuit design editor** (US 9,305,133 -- Apr 2016) -- Semi-transparent constraint editor for in-context violation reconciliation during layout editing, with simplified hierarchy visualization and lookup-order modification.
- Methods, Systems, and Computer-Program Products for Item Selection and Positioning Suitable for High-Altitude and Context Sensitive Editing of Electrical Circuits** (US 8,612,923 -- Dec 2013) -- High-altitude editing capabilities for congested layouts/schematics with context sensitivity, neighborhood awareness, and user-intent anticipation.
- System and method for process rules editors** (US 8,327,315 -- Dec 2012) -- Graphical editor for visualizing, modifying, creating, and removing process rules (constraints) through a GUI, with constraint groups associated to specific circuit design objects.

7. **System and method for aperture based layout data analysis to achieve neighborhood awareness** (US 8,271,909 -- Sep 2012) -- Input infrastructure that gathers information surrounding the cursor's locality, analyzes it in view of the issued command, and automatically determines suitable targets or next operations.

AWARDS & RECOGNITION

Award	Context
Organization Level Award (x2)	Virtuoso Layout Editor features and Constraint GUI
Excellence in Innovation	Novel approaches to layout automation
Excellence in Execution	Constraint GUI delivered from scratch
Cadence Encore Award	Outstanding technical contribution
Departmental Awards (x2)	LEF-DEF/GDS translator performance and quality

PUBLICATIONS

- Paper presented at **Cadence Technical Conference** (US) -- Constraint editing system design and implementation.
- Paper presented at **Cadence Technical Conference** (US) -- LEF-DEF/GDS translation performance optimization.

EDUCATION

Bachelor of Technology (B.Tech), Mechanical Engineering Indian Institute of Technology (IIT), Delhi -- Class of 2002

STRATEGIC CUSTOMER ENGAGEMENTS

Direct technical lead for critical EDA support and methodology deployments at:

Tier 1 Silicon: Apple, NVIDIA, Intel, Samsung, TSMC, Qualcomm **Tier 2 Silicon:** Broadcom, AMD, Sony, MediaTek, Amazon, Altera, Microsoft, Infineon, Renesas

References available upon request.